

SHRI SANJAYKUMAR V

Software Engineering Student | Cloud | Generative AI | Full-Stack Development

Vellore, India | +91 6369945011 | [Email](#) | [LinkedIn](#) | [GitHub](#)

Professional Summary

Software Engineering student pursuing an M.Tech Integrated in Software Engineering at Vellore Institute of Technology with a 9.12/10.0 CGPA. Hands-on experience in software engineering, cloud computing, enterprise Generative AI, RAG, vector search, machine learning, backend development, REST APIs, microservices, system design, CI/CD, and Agile delivery. Completed an academic internship focused on SAP Generative AI technologies, Python, SAP BTP, enterprise RAG workflows, vector search, and backend integration. Strong foundation in data structures and algorithms, object-oriented programming, databases, distributed systems, testing, debugging, and technical problem solving.

Education

Vellore Institute of Technology, Vellore

M.Tech Integrated in Software Engineering

Expected Graduation: 2028

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, Computer Networks, Operating Systems, Database Management Systems, Cloud Computing, Machine Learning, Statistics and Probability

Aug 2023 – Present

CGPA: 9.12/10.0

Technical Skills

- **Languages:** Python, Java, C++, C, JavaScript, TypeScript, SQL, PHP
- **Computer Science:** Data Structures and Algorithms, Algorithms, Graph Algorithms, Object-Oriented Programming, Design Patterns, Concurrency, Multithreading, Synchronization, System Design
- **Software Engineering:** Software Architecture, REST APIs, Microservices, Backend Development, SDLC, Agile, Scrum, CI/CD, Software Testing, Debugging, Git, GitHub
- **Cloud and DevOps:** Microsoft Azure, Azure Virtual Machines, Azure IaaS, Azure Network Security Groups, AWS EC2, Docker, Docker Compose, Linux, SSH, Cloud Infrastructure
- **Generative AI and Machine Learning:** Generative AI, Large Language Models, RAG, Retrieval-Augmented Generation, Vector Embeddings, Vector Search, Semantic Search, Prompt Engineering, OpenAI API, SAP Generative AI Hub SDK, scikit-learn, XGBoost, LSTM, Random Forest, Gradient Boosting
- **Web and Data:** React, Next.js, Node.js, Express.js, Firebase, MongoDB, Firebase Realtime Database, Tailwind CSS, Pandas, NumPy, Matplotlib, Chart.js
- **Enterprise Technologies:** SAP BTP, SAP ABAP Cloud, Jupyter, Enterprise AI Workflows, Backend Integration

Experience

TForce India Private Limited (Formerly Qcloud Tech Private Limited)

Academic Intern – Generative AI and SAP Technologies

May 2026 – July 2026

Chennai, India

- **Developed** enterprise Generative AI workflows using Python, Jupyter, SAP Generative AI Hub SDK, and SAP BTP concepts, strengthening practical experience with cloud-based AI application development.
- **Implemented** enterprise RAG workflows using vector search and retrieval-based context generation to support grounded AI responses and improve the relevance of domain-specific outputs.
- **Integrated** Generative AI components with backend application workflows, gaining hands-on exposure to enterprise service integration, API-driven development, and SAP ABAP Cloud.
- **Evaluated** AI workflow behavior through Python-based experimentation and iterative testing, applying software engineering practices to enterprise Generative AI development.

Projects

LEGITIFY – Security and Compliance Analysis Platform *Python / Application Security / DevSecOps / GitHub*

- **Engineered** a security-focused analysis platform to identify repository-level security and configuration risks, applying secure software development and DevSecOps principles.
- **Automated** policy-based analysis workflows to surface security and compliance issues across source-code management assets, improving repeatability of repository checks.

CampusLLM – Personalized University Assistant *Python / Next.js / React / RAG / Vector Search / OpenAI API*

- **Designed** a Retrieval-Augmented Generation system using vector embeddings, semantic search, and intelligent query routing to deliver source-cited responses from university documents.

- **Optimized** retrieval, web fallback, and context-aware generation, achieving approximately **3-second average latency** across the end-to-end response pipeline.

Solar Energy Forecasting – End-to-End ML Pipeline *Python / scikit-learn / XGBoost / LSTM / NASA POWER API*

- **Built** an end-to-end machine learning pipeline and benchmarked **5 models** including Linear Regression, Random Forest, XGBoost, Gradient Boosting, and LSTM using MAE and RMSE for systematic model selection.
- **Engineered** multivariate weather features from the NASA POWER API and automated **7-day forecasting**, covering data acquisition, preprocessing, model training, evaluation, and prediction.

FinTrack – Personal Finance and Group Settlement Platform *Python / React / Node.js / Firebase / REST APIs / Graph Algorithms*

- **Architected** a full-stack application with REST APIs and real-time synchronization across **5+ modules**, applying modular architecture and fault-isolation principles for maintainability.
- **Optimized** a graph-based debt-settlement algorithm for multi-user transactions and integrated the OpenAI API for spending-pattern analysis and natural-language financial insights.

Cloud-Based Scalable System Architecture *AWS EC2 / Microsoft Azure / Docker / Linux / Microservices / CI/CD*

- **Containerized** a MERN-stack real-time application with Docker Compose, decomposing frontend, backend, and MongoDB into isolated services to support consistent development and deployment environments.
- **Provisioned** AWS EC2 and Microsoft Azure virtual machines and configured security groups, Azure NSG rules, SSH authentication, and IaaS security controls for cloud infrastructure deployment.

Leadership

-
- **Vice Chairperson & Core Member, VIT Health Club** 2025 – Present
Directed execution of the Hack the Hospital hackathon with **200+ participants**, coordinating cross-functional teams, technical workshops, timelines, and deliverables.
 - **Sponsorship Lead, Riviera & Gravitas Tech Festivals, VIT** 2025 – 2026
Managed external stakeholder communication and negotiations, securing **INR 60+ lakhs** in non-monetary sponsorships and coordinating deliverables across two **1,000+ attendee** festivals.